Knowledge-Aided Automated Synthesis for Broadband Power Amplifier With Transformer-Coupled Resonators

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***Abstract*— This paper presents a knowledge-aided automated synthesis method to design broadband power amplifiers (PAs) integrated with transformer-coupled resonators (TCRs). Different from the traditional TCR-based PA designs that focus solely on broadband load-pull matching or TCR transimpedance, the intrinsic design knowledge of the proposed method utilizes the mechanism of joint-design between PA active circuits and TCR networks. By combining this knowledge with an optimization-oriented genetic algorithm (GA), an initial TCR solution that achieves low-ripple and broadband PA output response can be rapidly synthesized, in the sense that the superior initial population, not as random as that in traditional GA, can improve the GA convergence and reduce the algorithm iterations. Simulated results show that this work achieves about 4 times speed improvement in TCR synthesis than the traditional GA. The proposed method can efficiently synthesize the optimal TCR parameters for multi-stage PA to obtain the user-defined output power within a target bandwidth. As a proof of this work, a cascode-based PA for FMCW radar application is designed in 110-nm CMOS process, aiming to realize a peak 14-dBm output power and 50% 1-dB fractional bandwidth (FBW). The measured PA large-signal performance versus frequency shows a wide 1-dB BW for P_{sat} over 8.5-14.5 GHz (52.2% FBW) with 15.7-dBm peak value and 27% peak PAE_{max} at 11.5 GHz, demonstrating the efficacy of the proposed broadband synthesis mothod for PAs.**

***Index Terms*— Knowledge-aided, electronic design automation (EDA), genetic algorithm (GA), broadband, power amplifier (PA), transformer-coupled resonator (TCR), complementary metal oxide semiconductor (CMOS).**

I. Introduction

MINIATURIZED radar sensors have been extensively applied to both military and civil aspects. Along with the evolution of technology nodes, the mature CMOS processes present the benefit of high integration, low power consumption and cost-efficient [1], [2], [3], [4], whereby, facilitating the implementation of monolithic radar sensors. To guarantee precise target judgment and detection, the abundant radar applications, e.g., the frequency modulated continuous wave (FMCW) scenario, pose the requirement for high range resolutions. Since the range resolution is inversely proportional to the FMCW chirp bandwidth [5], [6], a wide operating bandwidth is necessary for the FMCW transceiver link. As one of the critical modules, power amplifier (PA) not only dominates the output power and efficiency of the transmitter, but also affects the available chirp bandwidth. Numerous efforts focused on broadband PAs have been developed. One of the common implementation methods is to use the transformer-coupled resonators (TCRs) [7], [8], [9], [10]. As a high-order passive matching circuit, TCR features multi-poles transimpedance in the frequency domain, which gets closer to the Bode-Fano limit than other equivalent-order filter-based resonators [11], and thus effectively expanding the continuous bandwidth of PA. Moreover, with the advantage of compact area, TCR facilitates the layout connections in multi-stage PA.

Due to the multiple component parameters, and the close correlation between the desired targets (i.e., input impedance, insertion loss and transimpedance), TCR design usually accompanies inefficient manual parameter tuning, which is also a common problem in other high-order passive matching networks. Computer-aided algorithm (CAA) can help to accelerate circuit design since they can automatically search for the globally-optimal parameters and thus reduce the human intervention [2], [12], [13], [14], [15], [16], e.g., genetic algorithm (GA) [2], [12], [13] and artificial neural network (ANN) synthesizer [14], [15]. Nevertheless, the overall efficiency of circuit synthesis is still limited by the randomness of GA initial population or the time-consuming training data preparation in ANN.

Design knowledge plays important roles in TCR implementation as they can provide insight into how TCR operates. Therefore, embedding knowledge into CAA can guide TCR synthesis in a more straightforward way, with less randomness and no need for training data preparation. However, many published design knowledge of the broadband TCR-based

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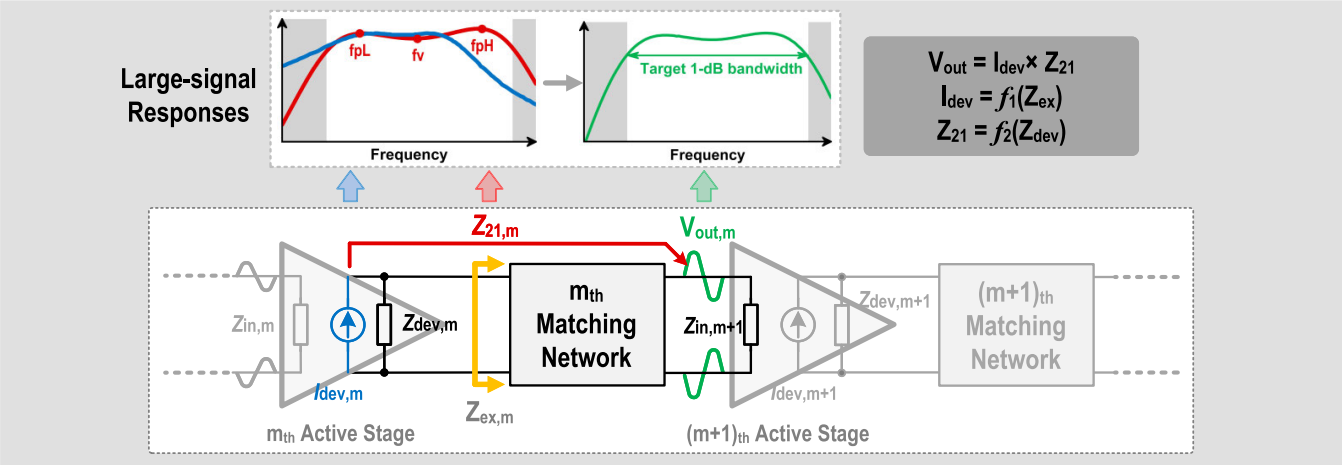


Fig. 1. Basic concept of the proposed design method for broadband multi-stage PA.

PAs are focused solely on PA load-pull matching [17], [18] or the broadband TCR transimpedance realization [7], [9]. The formers pay little attention to the power transmission bandwidth of the TCR networks, while the latters ignore the impact of TCR impedance on the active circuit's large-signal responses. All of them require additional tuning efforts on the PA output power bandwidth after cascading the active circuits and TCR networks, which degrade the design efficiency. For the TCR-based PA, its output power bandwidth is determined by the combined frequency responses of both active circuits and TCR networks. Interaction between PA active and passive parts should be focused on, especially for the multi-stage PA architecture.

In this paper, a knowledge-aided automated synthesis method (KASM) for broadband TCR-based PAs is introduced. At a given PA active topology, it can efficiently synthesize the realizable and optimal TCRs to obtain the user-defined output power within a target bandwidth. The intrinsic design knowledge is constructed by in-depth TCR impedance analysis (i.e., input impedance and transimpedance) and the mechanism of joint-design between the active stages and TCR networks. The proposed KASM consists of two major phases. Firstly, constraint rules and design guidance based on the constructed knowledge are used for rapid synthesis of an initial TCR that achieves low-ripple and broadband PA output response. Then, based on the initial superior TCR solution, an elitist genetic algorithm (a variant of NSGA-II [19]) is performed to further optimize the large-signal bandwidth of each PA stage in the next phase, where an intrinsic current extraction strategy assists to evaluate the large-signal characteristic. With the aid of knowledge, the proposed design method can dramatically reduce the randomness in TCR synthesis, which helps to improve the algorithm convergence and accelerate the TCR optimization. Therefore, the overall design efficiency of multi-stage PA is significantly improved.

To evaluate the effectiveness of this work, a cascode-based PA for FMCW radar application is designed via KASM in 110-nm CMOS process, aiming to realize a peak 14-dBm output power and 50% 1-dB fractional bandwidth (FBW). A traditional GA with a random initial population is deployed as a control group for comparison with KASM. It reveals that the KASM is more efficient in TCR synthesis since it requires about 4 times fewer iterations and running time. Meanwhile, the results obtained by KASM are more concentrated in distribution and closer to the user-defined target specification. The measurement results show that the fabricated PA chip exhibits a peak saturated output power of 15.7 dBm with 52.2% 1-dB FBW, which proves the efficacy of the proposed KASM for broadband PAs.

II. Analysis and Design Knowledge of TCR for Broadband PA

In this section, the basic principle of designing broadband multi-stage PA will be introduced. The TCR network, as a crucial module for achieving broadband response in this work, will be elaborated as an extension of [20], including the analysis of its transimpedance (Z_{21}) and input impedance (Z_{ex}) as well as their interaction with PA active stages. The presented analysis and considerations finally contribute to instructive design knowledge for the subsequent TCR automated synthesis method.

A. Principle of Broadband PA Operation

Broadband multi-stage PA requires the joint design of active and passive circuits for each stage. The basic concept of the proposed design method is illustrated in Fig. 1. In each stage, the bandwidth of the output voltage V_{out} is focused on. With a nearly constant driving voltage, the subsequent active stage is able to generate low-ripple output power within a target frequency range [21].

The bandwidth of each stage's V_{out} depends on the combination of the active stage's output current I_{dev} and the TCR network's transimpedance Z_{21}. Significantly, since PA is under large-signal operation, I_{dev} of each stage is load-dependent and thus the input impedance Z_{ex} introduced by the matching network would affect I_{dev}. Moreover, Z_{21}, which is defined as the transfer impedance between I_{dev} and V_{out}, is correlated with the device impedance Z_{dev} of the active stage. In other words, I_{dev} is the function of Z_{ex}, and Z_{21} is the function

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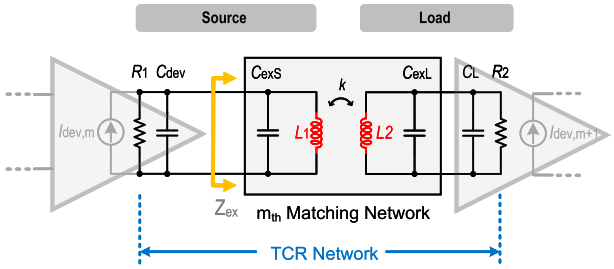


Fig. 2. Architecture of the TCR network.

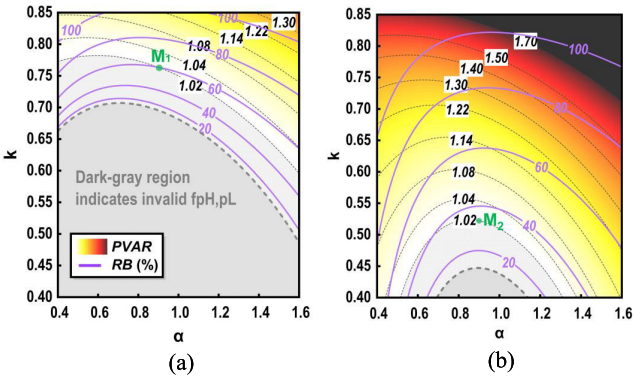


Fig. 3. Contour plots of the $PVAR$ and RB with different α, k and Q. (a) $Q = 1.0$. (b) $Q = 2.0$.

of Z_{dev}. The function relationship indicates the interaction between the active stage and the matching network.

To realize a broadband multi-stage PA with maximum output power at a determined active topology, it is necessary to extend the bandwidth of the driving voltage and to design the nearly optimal load impedance (i.e., $Z_{\text{ex}} \approx Z_{\text{opt}} = R_{\text{opt}}||(-1/j\omega C_{\text{dev}})$ [4], [21]) within a wide frequency range for each active stage. The considerations of the function effect, i.e., $I_{\text{dev}} = f_1(Z_{\text{ex}})$, $Z_{21} = f_2(Z_{\text{dev}})$, are also indispensable during the design process of the broadband multi-stage PA, and they will be discussed in Section II-D based on the subsequent analysis of Z_{21} and Z_{ex}.

B. Transimpedance Analysis of TCR Network

On-chip transformer-coupled resonators (TCRs) have been explored and widely applied in the amplifiers' matching networks. As shown in Fig. 2, the classical TCR network based on [7], [9], [11] consists of a magnetically coupled transformer and dual RC branches. On the source side, R_1-C_{dev} model the equivalent device impedance Z_{dev} of the active stage, while R_2-C_{L} on the other side model the equivalent input impedance of the load (e.g., Z_{in} of Fig. 1, or the G-S-G output pads). $\{R_{1,2}, C_{1,2}, L_{1,2}, k\}$ are the TCR basic parameters to be designed, where C_1 and C_2 represent $C_{\text{dev}} + C_{\text{exS}}$ and $C_{\text{L}} + C_{\text{exL}}$, respectively. In this work, $\{\omega_{1,2}, Q_{1,2}, \alpha\}$ are used as the intermediate variables to assist in TCR design, where $\omega_{1,2}$ are the resonant frequencies of $L_{1,2}C_{1,2}$ tanks, and the quality factors $Q_{1,2} = R_{1,2}/(\omega_{1,2}L_{1,2})$. The factor α is defined as the ratio of ω_1 to ω_2. Compared to the basic parameters, $\{\omega_{1,2}, Q_{1,2}, \alpha\}$ are more intuitive and they could provide better insight into TCR's performance [7], [22].

For the broadband PA, the TCR network's transimpedance normally features dual peaks and a single valley in the frequency domain. To obtain broadband power transmission, two peaks of $|Z_{21}|$ need to reach the similar amplitude. Typically, the equal-peaks condition is given by [22]

$$\frac{\omega_1}{\omega_2} = \frac{Q_1}{Q_2} = \alpha, \tag{1}$$

which can be further rewritten as

$$\frac{L_2}{L_1} = \alpha^2 \frac{R_2}{R_1}. \tag{2}$$

To achieve an in-band uniform $|Z_{21}|$ response, equation (1) merits further exploration. By combining (1) and the Z_{21} expression in [7], the peak and valley frequencies of Fig. 1 can be obtained by solving the partial differential of $|Z_{21}|$ with respect to the frequency, i.e., $\partial\,|Z_{21}(\omega)|\,/\partial\omega = 0$:

$$f_{\text{pL,pH}} = \frac{\omega_2}{2\pi}\sqrt{\frac{\varepsilon \mp \sqrt{\varepsilon^2 - 4\alpha^2 Q^4\left(1-k^2\right)}}{2Q^2\left(1-k^2\right)}} \tag{3}$$

$$f_{\text{v}} = \frac{\omega_2}{2\pi}\sqrt{\frac{\varepsilon + \sqrt{\varepsilon^2 + 12\alpha^2 Q^4\left(1-k^2\right)}}{6Q^2\left(1-k^2\right)}} \tag{4}$$

where $\varepsilon = \left(1+\alpha^2\right)Q^2 - \left(1-k^2\right)$ and $Q = Q_2$.

Then, the peak-to-valley amplitude ratio ($PVAR$), i.e., $\left|Z_{21}(f_{\text{pL,pH}})\right|/|Z_{21}(f_{\text{v}})|$, and the customized relative bandwidth (RB) can be deduced respectively according to (3) and (4):

$$PVAR(\alpha, k, Q) = \frac{|\delta|}{9Q^2}\sqrt{\frac{3(\delta+3\mu)}{\left(1-k^2\right)\left[1+\alpha^4-2\alpha^2\left(1-2k^2\right)\right]}} \tag{5}$$

$$RB(\alpha, k, Q) = \frac{f_{\text{pH}} - f_{\text{pL}}}{f_{\text{v}}} = \sqrt{\frac{6(\varepsilon - 2\alpha Q^2\sqrt{1-k^2})}{\varepsilon+\mu}} \tag{6}$$

where

$$\begin{cases} \delta = 2\varepsilon - \mu + 3(1-k^2) \\ \mu = \sqrt{\varepsilon^2 + 12\alpha^2 Q^4(1-k^2)}. \end{cases} \tag{7}$$

Based on (5)-(7), contour plots of $PVAR$ and RB with different α, k and Q can be constructed, as shown in Fig. 3. For a given α and taking the markers M_1 and M_2 for examples, it can be intuitively observed that 1) the coupling coefficient k associated with the same $PVAR$ contour (as well as the same RB contour) will decrease with increasing Q. It means that for the broadband TCR transimpedance design in high-Q case, the coupling of the transformer will not be tight, which may lead to increased power loss and impair the transformer efficiency [23]. 2) The RB associated with the same $PVAR$ contour will become smaller with increasing Q. Hence, in order to maintain the same in-band ripple of $|Z_{21}|$ in high-Q case, the relative bandwidth will be degraded.

The dark-gray regions in Figs. 3(a)-(b) indicate the inappropriate combinations of k and α that cause invalid $f_{\text{pH,pL}}$

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and extremely narrow bandwidth. Significantly, the dashed borderlines of these regions are determined by $\varepsilon^2-4\alpha^2Q^4(1-k_m{}^2)=0$ of (3), where the minimum coupling coefficient k_m to flatten the $|Z_{21}|$ response (i.e., $PVAR=1$) can be solved as the function of α and Q:

$$k_m=\sqrt{1-Q^2(1+\alpha^2+2\alpha^2Q^2)+2\alpha Q^3\sqrt{1+\alpha^2+\alpha^2Q^2}}. \tag{8}$$

In practical design, the choices of α and Q depend on not only R_2/R_1 and L_2/L_1 (according to (1) and (2)), but also the input impedance, which will be elaborated upon in the next part. Once the values of α and Q are chosen, the closed-form solution (8) can provide an accurate and expeditious means to determine k. Concurrently, the adoption of (8) can guarantee the in-band flatness of $|Z_{21}|$, even under the influence of active stage's output resistance, which will be explained in Section II-D.

C. Input Impedance Analysis of TCR Network

For the power matching of the active stage, the impedance Z_{ex} in Fig. 2 is desired to be $R_{opt}||(-1/j\omega C_{dev})$ according to [4] and [21]. In this section, simulations are conducted with the assumption of $R_1=R_{opt}$. To balance the in-band flatness of $|Z_{21}|$, equations (1) and (8) are also carried out in the following simulations.

To gain the insight, different parts (R_p, C_p) of the target admittance are explored respectively, where

$$R_p=1/\mathrm{Re}[Y_{ex}] \tag{9}$$

$$C_p=\mathrm{Im}[Y_{ex}]/\omega+C_{dev} \tag{10}$$

and $Y_{ex}=1/Z_{ex}$. Fig. 4 shows the impact of $\{\alpha, Q\}$ on R_p and C_p, which are normalized to R_{opt} and R_{opt} is the optimum parallel resistance for maximum output power:

1) *At a given Q*, Figs. 4(a)-(b) reveal that for $\alpha<1$, a lower α would cause R_p to deviate further from R_{opt}. While for $\alpha>1$, a larger α would cause C_p to be further away from the desired value of 0, and also lead to larger variation of C_p in the vicinity of f_v. The deviation of R_p or C_p would degrade the power matching.
2) *At a given α*, Figs. 4(c)-(d) reveal that the increase of Q would degrade the impedance matching between R_p and R_{opt}, and also result in larger variation of C_p.

In terms of broadband power matching, C_p variation is desired to be small, and high return loss for the matching of R_p and R_{opt} is also necessary. Therefore, $\alpha\approx1.0$ for high-Q cases (e.g., $Q\geq2.0$), or $\alpha\in[0.8, 1.2]$ for low-Q cases are chosen as the constraint rules in this work. Under these constraints, C_p variation can be guaranteed to be small with nearly 50% fractional bandwidth ($0.75f_v\sim1.25f_v$). Thus, C_p that deviates from zero at around f_v (for $\alpha\neq1$) can be compensated with the assist of capacitor C_{exS} in Fig. 2. The implementation of such capacitor assisting strategy (CAS) depends on the value of C_p at f_v. For $C_p<0$, C_{exS} should be increased to enlarge C_p. While for $C_p>0$, C_{exS} should be decreased.

In summary, when assuming $R_1=R_{opt}$, a decent return loss of the matching between R_p and R_{opt} can be achieved by the proper choices of Q and α. However, R_1 of the active stage is actually different from R_{opt}. Though the above analysis of the TCR's input impedance will not be affected, the influence of actual R_1 on TCR's $|Z_{21}|$ needs reexamination. Further discussions on the interaction between PA active stage and TCR network will be presented below.

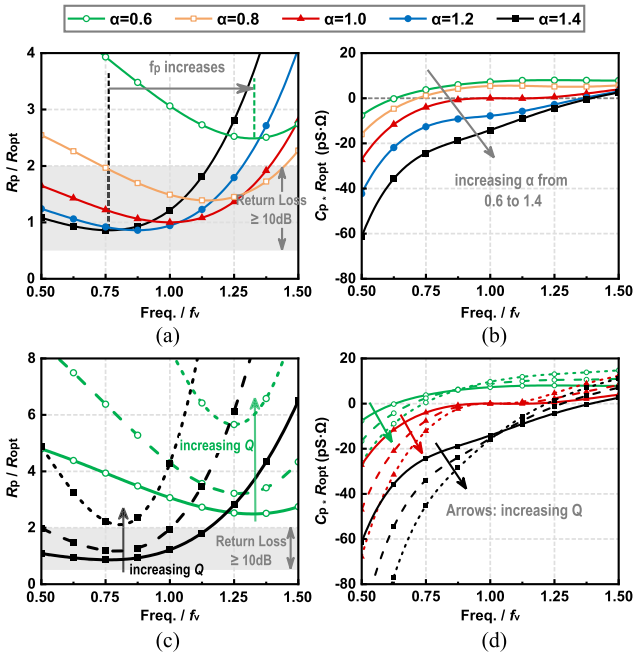


Fig. 4. Simulated results of (a) normalized R_p versus α with $Q=1.0$, (b) normalized C_p versus α with $Q=1.0$, (c) normalized R_p versus Q (1.0-2.0 with step of 0.5) with $\alpha=0.6$ and 1.4, and (d) normalized C_p versus Q (1.0-2.0 with step of 0.5) with $\alpha=0.6$, 1.0 and 1.4.

D. Interaction Between Active Stage and TCR Network

The interaction between active stage and load network is intrinsic in the PA. A design ignoring this interaction will bring negative effect on the overall bandwidth, especially when the number of PA stages increases. There are two critical parts of the interaction between the PA active stage and the TCR network. One is $I_{dev}=f_1(Z_{ex})$ and the other part is $Z_{21}=f_2(Z_{dev})$.

1) Impact of Z_{dev} on Z_{21}: At the given Q and α ($\alpha\neq1$), TCR network based on Equations (1), (8) and $R_1=R_{opt}$ features a flat $|Z_{21}|$ response within the frequency range of f_{pL}-f_{pH}. However, the difference between R_1 and R_{opt} would cause undesired ripple in $|Z_{21}|$ response due to the violation of (1), and ultimately affect the bandwidth of the output power. To tackle this issue, CAS is adopted, which provides an effective way to restore the TCR's $|Z_{21}|$ response. (Assuming $R_1>R_{opt}$) Its implementation and effect depend on two different cases of α:

- If $\alpha<1$, adjust the R_1C_1 product by decreasing C_{exS}. The new value of C_1 is determined by R_2C_2/R_1. In this way, equation (1) is satisfied and the flatness of $|Z_{21}|$ can be regained. Such benefit is closely related to (8) under the condition of $R_1=R_{opt}$. Note that the decrease of C_{exS} will lead to the increase of ω_1/ω_2, and thus

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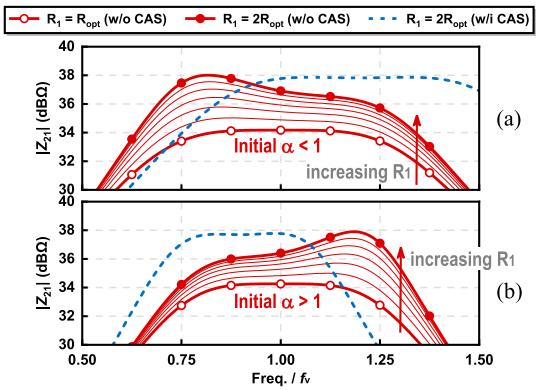


Fig. 5. The impact of increased R_1 on initial $|Z_{21}|$, and the optimization effect of CAS. (a) Initial $\alpha < 1$. (b) Initial $\alpha > 1$.

larger α. From (4), the valley frequency (f_v) of $|Z_{21}|$ will become higher accordingly. An example for illustration is presented in Fig. 5(a), which takes $\alpha = 0.8$ as an initial design parameter for the TCR network. The increase of R_1 causes $|Z_{21}|$'s left peak to become higher. With the adoption of CAS, a new in-band uniform $|Z_{21}|$ response with a higher f_v is achieved.

- If $\alpha > 1$, adjust L_1C_1 by increasing C_{exS}, and the new C_1 is determined by L_2C_2/L_1. Likewise, equation (1) can be satisfied, resulting in an in-band uniform $|Z_{21}|$ response. Moreover, α is decreased to 1 after adopting CAS. The change of α will reduce the f_v of $|Z_{21}|$. The illustration of this case ($\alpha = 1.2$) is presented in Fig. 5(b).

In the above cases, the adjustment trend of C_{exS} is exactly the same as the Y_{ex} optimization case in Section II-C, which means that $|Z_{21}|$ and the susceptance of Y_{ex} can be optimized simultaneously by CAS. Moreover, due to the f_v offset caused by CAS, f_v and the center frequency (f_c) of the PA operating band should be staggered according to the initial value of α ($\alpha \neq 1$), which can be summarized below:

$$\begin{cases} case\ 1: \text{ if } \alpha < 1, \text{ then set } f_v < f_c \\ case\ 2: \text{ if } \alpha > 1, \text{ then set } f_v > f_c \end{cases} \tag{11}$$

2) Impact of Z_{ex} on I_{dev}: To gain the insight, an intrinsic current extraction strategy is firstly deployed to capture the large-signal characteristic of the PA active stage. Its operation concept and brief algorithm are shown in Fig. 6(a) and Algorithm 1, respectively. The input variable R_p at different design frequencies is obtained through the simulations in Section II-C. Then, by Algorithm 1 and the following calculation of

$$|I_{dev}| = \sqrt{2P_L R_p \left[\left(\frac{1}{R_p}+\frac{1}{R_1}\right)^2 + (2\pi f_i C_{dev} + \mathrm{Im}[Y_{ex}])^2\right]}, \tag{12}$$

the amplitude of the intrinsic output current at different design frequencies can be obtained.

A cascode-based power stage of our fabricated PA is used as an illustrative example. Its related component parameters will be presented in Section IV, which will not be elaborated here. Figs. 6(b)-(c) present the input variables $R_p(\boldsymbol{f})$ with

Algorithm 1 Intrinsic Current Extraction Strategy

Variables:
$\boldsymbol{f}$: = [f_1, f_2, …, f_m], design frequencies
R_p: Parallel resistance of Z_{ex} at each frequency
$|I_{dev}|$: Amplitude of the output current at each frequency
P_L: Output power at each frequency

Input: $R_p(\boldsymbol{f})$
Output: $|I_{dev}(\boldsymbol{f})|$
1: Load-pull simulation to obtain C_{dev} at f_v
2: $\mathrm{Im}[Y_{ex}] = -2\pi f_v C_{dev}(f_v)$ according to CAS
3: **for** i = 1: m **do**
4: Load-pull simulation to obtain C_{dev} at f_i
5: Periodic steady state simulation to obtain P_L at f_i
6: Periodic steady state simulation to obtain Z_L at f_i
7: $R_1 = 1/\mathrm{Re}[1/Z_L]$
8: Calculate $|I_{dev}|$ using Equation (12)
9: **end for**
10: **return** $|I_{dev}(\boldsymbol{f})|$

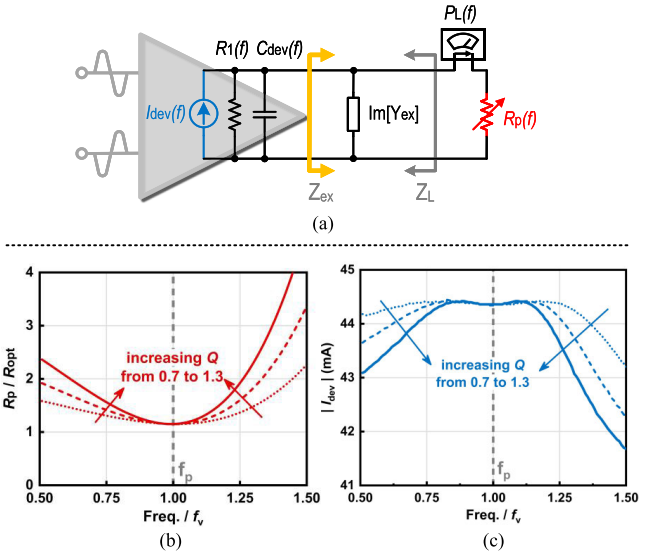


Fig. 6. (a) Operation concept of the intrinsic current extraction strategy. (b) Normalized $R_p(\boldsymbol{f})$ and (c) calculated $|I_{dev}(\boldsymbol{f})|$ responses versus Q with $\alpha = 0.9$ (where Q = 0.7, 1.0 and 1.3).

different Q and the corresponding $|I_{dev}(\boldsymbol{f})|$ of the power stage based on Algorithm 1. The load (Z_{ex})-dependent characteristic of the intrinsic output current can be clearly observed. In detail, the $|I_{dev}(\boldsymbol{f})|$'s bandwidth can be controlled by $R_p(\boldsymbol{f})$ responses with different Q. Significantly, the center frequency of $|I_{dev}(\boldsymbol{f})|$ is dominated by the valley frequency (f_p) of $R_p(\boldsymbol{f})$. In this work, f_p is typically set to the PA center frequency, i.e., $f_p = f_c$.

By solving the partial differential of R_p with respect to the frequency, f_p can be given by

$$f_p = \frac{\omega_2}{2\pi}\sqrt{\frac{1}{1-k^2} - \frac{1}{2Q^2}}. \tag{13}$$

Combined with (4), f_p can be expressed as a function of α, Q, k and f_v. Setting $k = k_m$ based on (8), Fig. 7 depicts the

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Algorithm 2 Proposed KASM

Definition:
f_c: Center frequency of the operating band
$Lratio_{min}$, $Lratio_{max}$: = Min, max constraint of L_2/L_1
$\omega_{srf1,2}$: Self-resonance frequencies of $L_{1,2}$
$BW_{1\text{-dB}}$: Target 1-dB bandwidth
g_{max}: Maximum number of GA iterations

Input: f_c, $BW_{1\text{-dB}}$, $Lratio_{min,max}$, g_{max}, $R_{1,2}$, C_L, load-pull results R_{opt}, C_{dev}
Output: TCR parameters $L_{1,2}$, C_{exS}, C_{exL}, k
First Phase (Initial TCR generation):
1: Initialization: $k_m = 0.6$, $k_{step} = 0.05$
2: **while** 1 **do**
3: $\alpha = 1.2 - 0.4 \times rand$(1)
4: Calculate L_2/L_1 using Equation (2)
5: **if** $L_2/L_1 \in [Lratio_{min}, Lratio_{max}]$ **then break**
6: **end if**
7: **end while**
8: **while** 1 **do**
9: $k \leftarrow k_m$
10: Obtain Q using Equation (8)
11: Obtain f_v using Equation (13)
12: Calculate $L_{1,2}$, $C_{1,2}$ based on α, Q, k and f_v
13: **if** $\omega_{srf1,2} \geq 2\times$(maximum operating freq.) **then break**
14: **else**
15: $k_m \leftarrow k_m - k_{step}$
16: **end if**
17: **end while**
18: **if** $\alpha \neq 1$ **then**
19: Adopt CAS to optimize TCR's $|Z_{21}|$ and Z_{ex}
20: Update $C_{exS} \leftarrow C_{dev}(f_c) + \mathrm{Im}[Y_{ex}]/2\pi f_c = 0$
21: **end if**

Second Phase (Bandwidth optimization):
22: **for** i = 1: g_{max} **do**
23: Adopt elitist GA to optimize the bandwidth
24: **end for**
25: **if** $BW_{1\text{-dB}}$ is not satisfied **then**
36: Decreases the resistance ratio (R_1/R_2 or R_2/R_1)
27: Return to the First Phase
28: **end if**
29: **return** $L_{1,2}$, C_{exS}, C_{exL}, k

linear relationship between f_p/f_v and α with different Q, and indicates how to coordinate f_v and α to obtain the desired f_p. That is, selecting $f_v < f_p$ when $\alpha < 1$, or $f_v > f_p$ when $\alpha > 1$, which is consistent with the frequency staggering operation of (11). Therefore, the center frequencies of $|I_{dev}|$ and $|Z_{21}|$ can be designed and optimized to simultaneously approach f_c. It should be noted that f_p/f_v is not sensitive to Q and thereby f_p is dominated by f_v and α.

III. Proposed Automated Synthesis Method

Based on the the analysis and design knowledge of TCR, a comprehensive knowledge-aided automated synthesis method (KASM) can be developed for broadband PA. Its brief algorithm is presented in Algorithm 2 and the design procedure of each phase is given as follows.

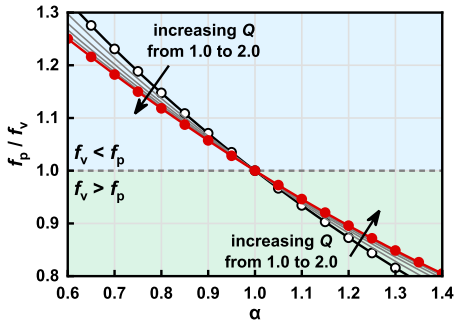


Fig. 7. The relationship between f_p and f_v with different α and Q.

1) *First Phase*: With the designed active stages, the KASM input parameters $\{R_{1,2}, C_L, R_{opt}, C_{dev}\}$ can be obtained through large-signal load-pull and steady-state simulations. Then, based on the assumption of $R_1 = R_{opt}$, KASM selects α within the range of 0.8∼1.2 to obtain a reasonable L_2/L_1.

With $k = k_m$ ($k_m = 0.6$ is selected initially), two solutions of Q can be determined by (8). The positive solution is retained and the non-positive one is discarded. If Q >2.0, then α will set to 1.0. Next, with the aim of $f_p = f_c$, f_v can be obtained by (13) and the initial solution of $\{L_{1,2}, C_{1,2}\}$ can be calculated accordingly. The transformer layout generation based on $L_{1,2}$ will be illustrated in Section IV-B. The self-resonance frequencies ($\omega_{srf1,2}$) of $L_{1,2}$ are evaluated by the electromagnetic simulation tool, which should be 2∼3 times of the PA's maximum operating frequency. Otherwise, k_m needs to be reduced and $\{L_{1,2}, C_{1,2}\}$ should be redefined through *step* 9-16. Finally, with the designed TCR network, CAS is adopted to optimize $|Z_{21}|$ (based on actual R_1) and the input impedance Z_{ex} simultaneously.

2) *Second Phase*: In this phase, an elitist genetic algorithm (GA) is performed to further optimize the TCR's output voltage ($|V_{out}|$ in Fig. 1) bandwidth and the Z_{ex} matching bandwidth under the given g_{max} limitation. The initial population for GA is created from the generated results $\{\alpha, Q, f_v, C_{exS,exL}\}$ in the first phase. During the GA operation, Algorithm 1 is utilized to extract the $|I_{dev}|$ response, and $|V_{out}|$ is evaluated by the combined responses of $|Z_{21}|$ and the extracted $|I_{dev}|$. Besides, the input $Lratio_{min,max}$ remains a constraint in GA to ensure that the synthesized $L_{1,2}$ and their $\omega_{srf1,2}$ are reasonable. Finally, if the 1-dB bandwidth satisfies the user-defined target, the TCR parameters could be output. Otherwise, it indicates the improper resistance ratio (R_1/R_2 or R_2/R_1) of the TCR, and such ratio should be reduced by adding the source degradation inductors or parallel resistor [9], [24] in the PA active stage.

Considering the non-ideal effects in the implemented TCR network, such as the active stage's R_1 being a variant of frequency and the introduction of transformer parasitic resistors, they will cause uneven decrease in the $|Z_{21}|$ near the center frequency, thereby deteriorating the synthesized

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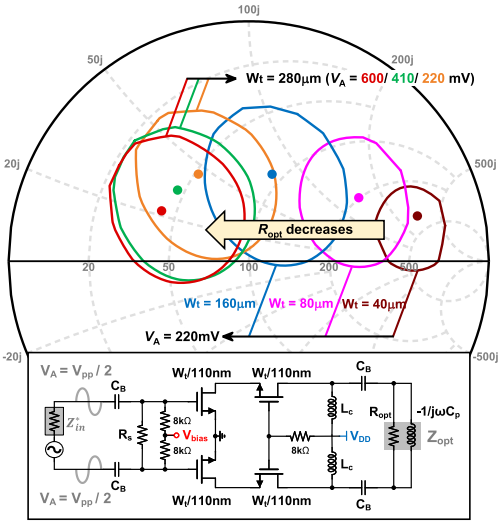


Fig. 8. Schematic of the pseudo differential cascode cell, and output power contours (points for $P_{\text{out,max}}$ while circles for $P_{\text{out,max}}$-1 dB) based on load-pull simulations at 12 GHz with $V_{\text{bias}} = 0.75$ V.

$|V_{\text{out}}|$ bandwidth from the proposed KASM. Therefore, before algorithm synthesis, it is necessary to set a certain margin on the user-defined bandwidth target, usually 10% $\sim$ 20%.

IV. Broadband PA Implementation

In order to validate the efficacy of the proposed KASM, a two-stage broadband PA for FMCW radar application is designed in Global Foundry 110-nm 1P7M CMOS process as an example, which aims to achieve a peak output power of 14 dBm at $f_c = 12$ GHz with a 1-dB fractional bandwidth (FBW) of 50%. In this work, pseudo-differential cascode cell shown in Fig. 8 is adopted for each PA active stage. The transistor width (W_t) of common-source and common-gate device is identical. A shunt resistor R_s is connected at the input [4] for circuit stability improvement and TCR resistance ratio (R_1/R_2) tuning. Besides, the drain supply voltage is 1.2 V in normal operation and temporarily fed through the RF chock inductors L_c, which will subsequently be replaced by the differential inductor of the TCR network. In this section, the active stage configuration, i.e., input voltage amplitude (V_A) and W_t, will be discussed first, which contributes to the practical design of the driver and power stage. Afterwards, the power stage with TCR-based output balun and the driver stage with TCR-based interstage matching network will be presented based on the back-to-front design procedure.

A. Active Stage Configuration

To gain an insight into the active stage configuration, load-pull simulations are carried out at 12 GHz with a certain gate bias (e.g., $V_{\text{bias}} = 0.75$ V, biasing the cascode cell at the peak-$f_{\max}$ current density). As shown in Fig. 8, the optimal impedance Z_{opt} is affected by both W_t and V_A, and given V_A and W_t, Z_{opt} can be determined and denoted as $Z_{\text{opt}}|_{V_A,W_t}$. Fig. 9(a) and (b) presents the power-PAE responses of the cascode cell at 12 GHz with different $Z_{\text{opt}}|_{V_A,W_t}$ to further discuss the impact of W_t and V_A.

Given $V_A = 220$ mV and different W_t (40, 80, 160, 280 μm), $Z_{\text{opt}}|_{220\text{ mV},W_t}$ can be obtained, respectively. After stabilizing by a proper R_s, the cascode cells with different W_t are connected to their respective $Z_{\text{opt}}|_{220\text{ mV},W_t}$. By varying the injected power, Fig. 9(a) shows the power-PAE responses of the cascode cells. All the marked symbols refer to the operating states whose input voltage is 220 mV. The operating single-ended output power ($P_{\text{os,out}}$) of these states increases with increasing W_t, but the operating power gain (PG) is nearly constant. Moreover, $P_{\text{os,out}}$ and DC power consumption are increased by the same order with increasing W_t in this case, and thus no obvious change in PAE (= η(1-1/PG) [25], where η is the drain efficiency).

Similar simulations are performed to explore the impact of V_A. Given $W_t = 10 \times 7\mu\text{m} \times 4$, Fig. 9(b) shows the simulated power-PAE responses of the cascode cells based on different $Z_{\text{opt}}|_{V_A,280\text{ um}}$ and the corresponding Z_{in}^*. The marked four kinds of symbols refer to the operating states whose input voltage is 110, 220, 410 and 600 mV, respectively. As input voltage increases, the $P_{\text{os,out}}$ of these states accordingly increases while the small-signal power gain reduces. This means that for the cascode cell of the same width, a higher input voltage is beneficial for more output power while maintaining a proper operating PG. For example, $P_{\text{os,out}}|_{220\text{ mV}} > P_{\text{os,out}}|_{110\text{ mV}}$ and $\text{PG}|_{220\text{ mV}} = \text{PG}|_{110\text{ mV}}$. Moreover, the operating PAE increases with increasing input voltage (when $\leq$ 410 mV). The higher input voltage ($>$ 600 mV), however, will lead to the degeneration of the operating PAE since severe power compression deteriorates the power gain.

The design procedures for the active stage configuration are summarized in Fig. 9(c). Driven by a moderate $V_A = V_a$, the cascode cell with small transistor width ($W_t = W_1$) initially operates near 1-dB compression point. Firstly, the increase of W_t within the power consumption requirement contributes to higher operating P_{out}. During this process, R_{opt} decreases but the operating power gain and PAE almost remain unchanged, as illustrated in Fig. 9(a). Secondly, increasing V_A to improve the operating P_{out}, PAE, and to further adjusts R_{opt}. To guarantee PA's operating bandwidth as well as the realization of proper TCR's transformer, R_{opt} is constrained within a range of no more than 40% of TCR's R_2. On this basis, the operating P_{out} and PAE can be further optimized by tuning W_t and V_A. The detailed flow chart for the cascode cell design is presented in Fig. 10.

B. Power Stage With Output Balun

In order to meet the output power requirement, a power margin of 1-2 dB needs to be reserved when considering the insertion loss caused by passive components. Therefore, the power stage with $W_t = 280$ μm (10 $\times$ 7 μm $\times$ 4 fingers) and V_A of 600 mV are selected based on the flow chart in Fig. 10. From Fig. 9(b), the $P_{\text{os,out}}$ at 12 GHz is 13 dBm, i,e,. 16-dBm differential power, and the operating PAE reaches the maximum value of 37.4%. With the considerations of balun's footprint and its connection, each transistor in power stage is separated into double parallel transistors with 5 $\times$ 7 μm $\times$ 4 fingers, which is more convenient for routing among the

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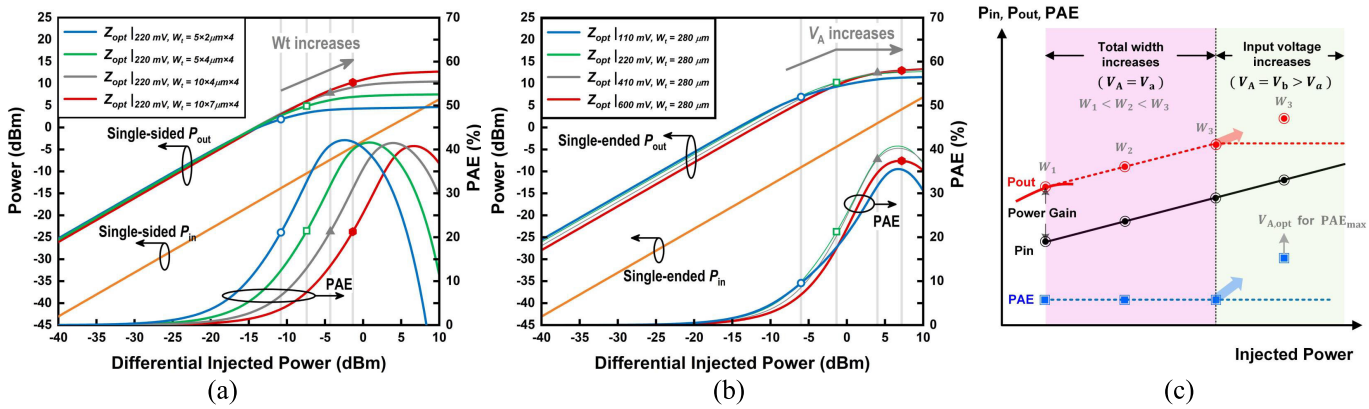


Fig. 9. Simulated power-PAE responses of the cascode cells with different loads at 12 GHz. (a) Loads are $Z_{\text{opt}}|_{220\text{ mV},W_t}$, where W_t = 40, 80, 160, 280 μm, respectively. (b) Loads are $Z_{\text{opt}}|_{V_A,280\text{ um}}$, where V_A = 110, 220, 410, 600 mV, respectively. (c) Summarized design procedures for the active stage configuration.

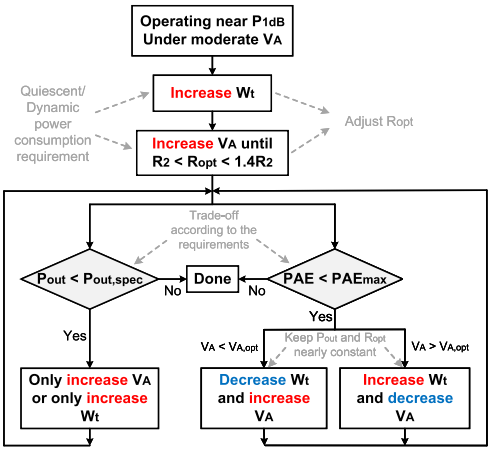
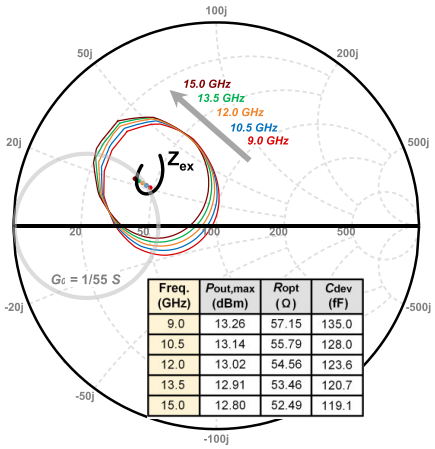


Fig. 10. Design flow chart for the basic pseudo-differential cascode cell.



Freq. (GHz)	Pout,max (dBm)	Ropt (Ω)	Cdev (fF)
9.0	13.26	57.15	135.0
10.5	13.14	55.79	128.0
12.0	13.02	54.56	123.6
13.5	12.91	53.46	120.7
15.0	12.80	52.49	119.1

Fig. 11. Single-ended output power contours (points for $P_{\text{out,max}}$ while circles for $P_{\text{out,max}}-1$ dB) of the power stage based on load-pull simulations (9-15 GHz), and the input impedance Z_{ex} (9-15 GHz) of the designed output balun.

active devices and output balun. Owing to the reduction of the gate parasitics [17], the power stage with such layout scheme obtains the highest operating power gain (G_{p}) compared to other schemes of the same W_{t}.

At a given differential injected power of 7.2 dBm, Fig. 11 shows the single-ended P_{out} contours of the power stage from 9 to 15 GHz. The Z_{opt} for $P_{\text{out,max}}$ at different input frequencies forms a locus that basically follows an equal-conductance circle (i.e., G_0 = 1/55 S), indicating that R_{opt} is frequency-independent and $R_2 < R_{\text{opt}} < 1.4R_2$, where R_2 = 50 Ω. Meanwhile, the capacitance C_{dev}, which can be derived from the load-pull results, slightly fluctuates within the operating frequency range. Thus, the TCR-based output balun needs to provide a resistance of about 55 Ω in parallel with an equivalent inductive impedance (i.e., $-1/j\omega C_{\text{dev}}$) over 9-15 GHz for the power stage.

Based on the confirmed active stage, KASM is deployed to automatically synthesize the required TCR parameters. In the first phase of Algorithm 2, an initial solution for $\{\alpha, Q, f_{\text{v}}, C_{\text{exS,exL}}\}$ is determined as {0.9, 1.3, 11.5, 153 fF, 382 fF}, and this knowledge-aided solution is then passed on to the next phase to accelerate the genetic algorithm (GA) in bandwidth optimization.

During the second phase, the efficacy of the proposed KASM will be demonstrated by a comparative simulation. In detail, an elitist GA with a random initial population is used as a control group for comparison with KASM (i.e., the core group). The core and control group share the same conditions, i.e., the number N of individuals per generation is 200, and the bound of GA variables $\{\alpha, Q, f_{\text{v}}, C_{\text{exS,exL}}\}$ is set to [0.5∼2.0, 0.5∼3.5, $0.75f_{\text{c}}$ ∼$1.25f_{\text{c}}$, 10 fF∼1 pF]. Fig. 12(a) shows the initial population of the core and control group, respectively. The frequency responses of balun's $|V_{\text{out}}|$ and Z_{ex} are both the GA optimized objects, which can be further subdivided and formulated as

$$\begin{cases} \text{min.}\,|obj_1| = \text{min.}\,|\text{FBW} \times \text{Center Value} - T_{\text{A}}| \\ \text{min.}\,|obj_2| = \text{min.}\,|\text{Center frequency} - f_{\text{c}}| \\ \text{min.}\,|obj_3| = \text{min.}\,|\text{Difference between dual peaks}|\,, \end{cases} \tag{14}$$

where T_{A} is the user-defined FBW×Center Value. According to the PA design goals, $T_{\text{A}} = 1.2\times50\times1.585$ for $|V_{\text{out}}|$ while $0.8\times50\times2.0$ for Z_{ex}-related VSWR, where 1.2 and 0.8 are both the scaling factors for compensating the non-idea effects

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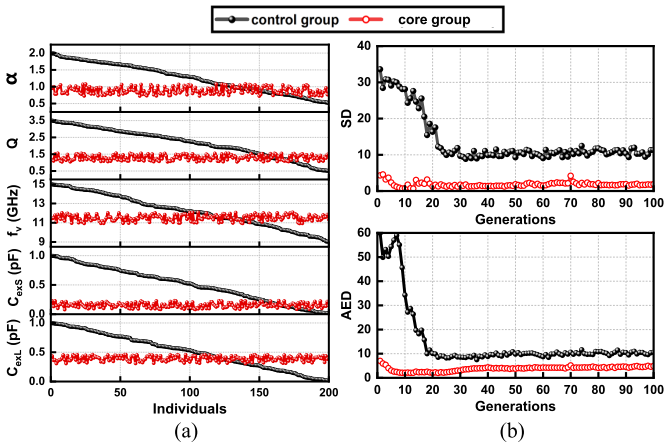


Fig. 12. (a) Initial population and (b) SD and AED in each GA generation of the core group (with knowledge aid) and the control group (without knowledge aid) respectively. Notes: 1) To clearly show the control group's initial population, each individual has been sorted according to its variable value. 2) The core group's initial population is generated by randomly adding offset values ($\leq 5\%$) on the determined knowledge-aided solution.

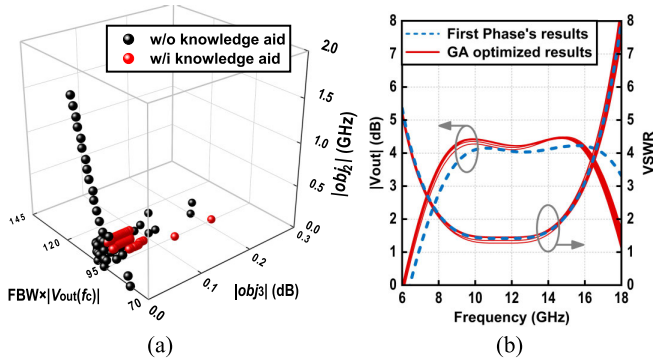


Fig. 13. (a) Comparison of the $|V_{\text{out}}|$ results between the core (with knowledge aid) and the control group (without knowledge aid) after 10 generations of GA operation. (b) The synthesized results after KASM's first phase, and the results of the core group after 10 generations of GA operation.

as demonstrated in Section III. To evaluate the optimized results of each GA generation, the difference between these results and the desired targets can be characterized by using the average Euclidean distance (AED):

$$\text{AED} = \frac{1}{M}\sum_{i=1}^{M}\sqrt{obj_{1,i}^2 + obj_{2,i}^2 + obj_{3,i}^2}, \tag{15}$$

where M represents the number of individuals in each generation that satisfy the constraint. As shown in Fig. 12(b), the AED value and the statistical standard deviation (SD) of the core group start to converge in the 5th generation, which are about 4 times speed improvement than the control group. Moreover, owing to lower AED and SD, the synthesized core group is much closer to the design goals. A more intuitive comparison is shown in Fig. 13(a), where the optimized results of the core group and the control group are presented respectively after 10 generations of GA operation. It can be clearly seen that the core group is more concentrated around the expected coordinates of $(T_{\text{A}}, 0, 0)$, which reveals that our proposed KASM can obtain the nearly optimal TCR parameters without much GA iterations in the second phase, and thus less running time. For KASM in this work, the maximum number of GA iterations is set to 10. Accordingly, the synthesized $|V_{\text{out}}|$ and Z_{ex}-related VSWR are shown in Fig. 13(b). Compared to the first phase's results, $|V_{\text{out}}|$'s center frequency optimized by GA is more closer to 12 GHz, while the Z_{ex}-related VSWR responses are almost unchanged. The optimized Z_{ex} on Smith Chart is plotted in Fig. 11.

Next, the TCR basic parameters $\{L_{1,2}, C_{1,2}, k\}$ can be determined from one of the KASM results, which is 797 pH, 576 pH, 282 fF, 376 fF and 0.61, respectively. In this case, the coplanar interwound layout is chosen as our balun structure due to its higher self-resonance frequency and ease of using thicker metals for both coils compared to the stacked layout [26]. In layout construction, the TCR model used in KASM [Fig. 14(a)] needs to be transformed into a more comprehensive model [Fig. 14(b)], which comprises capacitive coupling (C_m) and practical self-inductances (L_p, L_s). This transformation uses the concept of equivalent two-port Z-matrix, which generates the relationship among $L_{1,2}$ and $L_{p,s}$. The fitting relationships between C_m and k, produced by simulations on a larger number of transformer layouts with different diameters (D), are introduced to efficiently solve $L_{p,s}$ without mathematical iterations. After this, the primary and secondary coil are independently designed based on the calculated $L_{p,s}$. This can be achieved by using the inductance synthesis tool included in the PDK. In the last step, the coils are combined to form a transformer, as shown in Fig. 14(c). The 3-μm thick top metal layer (M7) is mainly used as the windings of the primary and secondary coils, while the lower metal layers (M4-M6) are used for underpass lines and the central tap connection. Fig. 14(d) shows the electromagnetic (EM)-simulated results of the designed balun.

The source ports on the balun's primary side are connected to the power stage and operated differentially. Fig. 15 shows their cascaded circuits by a simplified subgraph. From 9 to 15 GHz, the simulated insertion loss (i.e., power gain from differential C-D to E) is less than 1.32 dB. When the differential input voltage amplitude is 2 × 600 mV, the whole stage operating in compression state has the power gain of 5.14-6.48 dB, while the peak output power at 50-Ω load is about 14.65 dBm with the 1-dB bandwidth of over 6 GHz.

For the following driver stage design, the input impedance of the power stage merits attention. Under the fixed input voltage, there is a trade-off between power gain and stability, since they show opposite trends with the change of R_{s} (i.e., the resistor in Fig. 8). R_{s} is 130 Ω in this design, which not only ensures the unconditional stability and decent power gain, but also reduces the input parallel resistance of the power stage. The simulated equivalent input impedance of the power stage is around 110 Ω in parallel with a 170-fF capacitance at 9-15 GHz, offering a basis for the broadband interstage matching network design.

C. Driver Stage With Interstage Matching Network

According to Fig. 15, the power stage requires at least 5-dBm single-ended driving power. In circuit implementation, the driver stage with high power gain is attractive due to the PAE improvement of the overall PA. Based on the discussions

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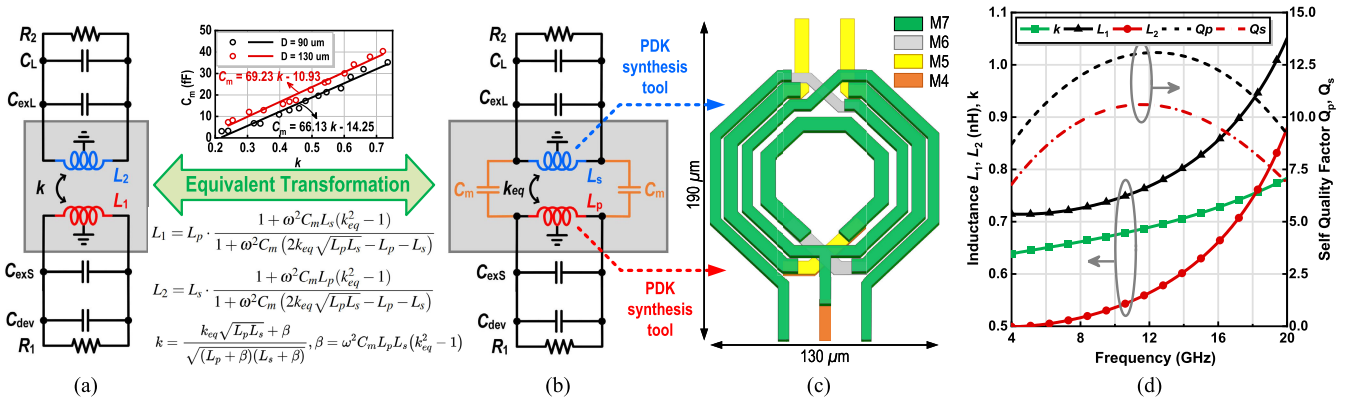


Fig. 14. (a) TCR model in KASM. (b) TCR model with capacitive coupling (C_m). (c) Transformer layout of the output balun and its (d) EM-simulated performance.

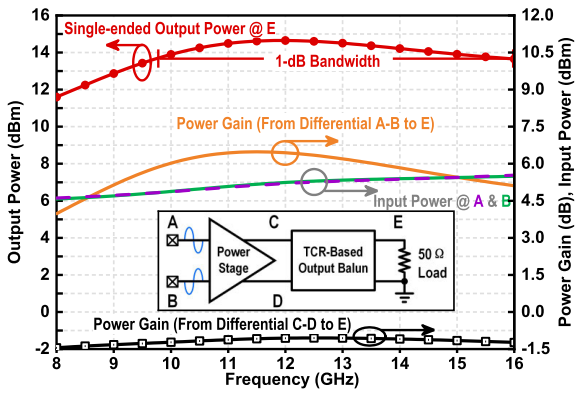


Fig. 15. Simulated power and power gain versus frequency for the power stage with TCR-based output balun.

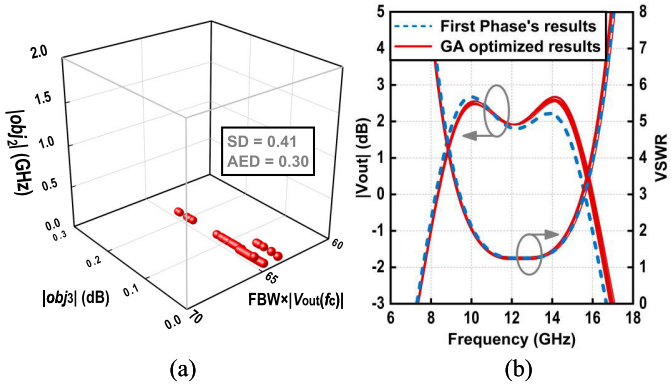


Fig. 16. (a) Optimized $|V_{\text{out}}|$ results of the IMN after 10 generations of GA operation. (b) The synthesized results of the IMN after KASM's first phase, and the results after 10 generations of GA operation.

of Fig. 9(b) and flow chart in Fig. 10, the driver stage with $W_{\text{t}} = 160\ \mu\text{m}$ ($10 \times 4\ \mu\text{m} \times 4$ fingers) and V_{A} of 220 mV is finally selected when R_{opt}, power gain and $P_{\text{os,out}}$ are focused on. As shown in Fig. 9(a), $P_{\text{os,out}}$ at 12 GHz is 7.87 dBm, which is higher than expected and reserves enough power margin when considering the insertion loss of the interstage matching network (IMN).

The following design process of the TCR-based IMN is similar to the output balun:

1) At a given differential injected power of −4.3 dBm [according to Fig. 9(a)], large-signal load-pull simulations are carried out to acquire the driver stage's R_{opt} and C_{dev} within 9-15 GHz, where $R_{\text{opt}} \approx 148\ \Omega$ and $C_{\text{dev}} \approx 77$ fF.

2) Based on the determined active stage, the first phase of KASM is performed to get an initial solution of $\{\alpha, Q, f_{\text{v}}, C_{\text{exS,exL}}\}$, which is {1.05, 2.0, 12.5, 153 fF, 80 fF}.

3) In the second phase of KASM, the user-defined T_{A} is set to 65 for IMN's $|V_{\text{out}}|$ while 80 for Z_{ex}-related VSWR according to the design goals. It should be noted that $|V_{\text{out}}|$ here represents the amplitude of IMN differential output voltage. Then, elitist GA is adopted to optimize both $|V_{\text{out}}|$ and VSWR. These optimized results after 10 generations of GA operation are shown in Fig. 16.

4) Finally, $\{L_{1,2}, C_{1,2}, k\}$ is determined from one of the KASM results, which is 941 pH, 757 pH, 214 fF, 251 fF and 0.43, respectively.

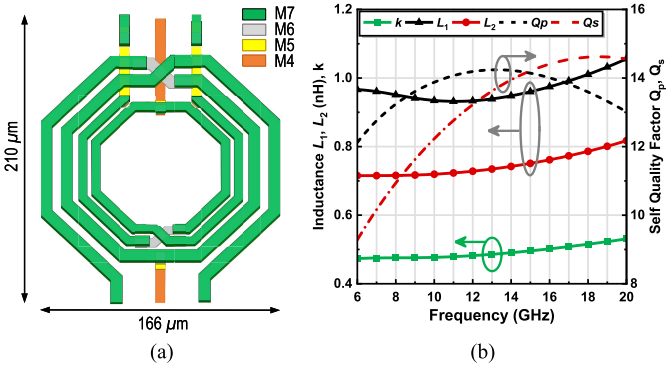


Fig. 17. (a) Transformer layout of the interstage matching network and its (b) EM-simulated performance.

In this case, the concentric coplanar layout [26] is adopted for the IMN transformer and Fig. 17 shows the transformer layout and the corresponding EM-simulated performance.

The primary and secondary ports of the IMN transformer are connected to the driver and power stage, respectively. The cascaded circuit of the two-stage PA is simplified as the subgraph of Fig. 18. Post-layout simulations reflect that when the differential input voltage amplitude of the driver stage is 2×220 mV, the output power of the IMN (i.e., at node A or B) peaks at 12 GHz and exceeds 5 dBm with a 1-dB bandwidth

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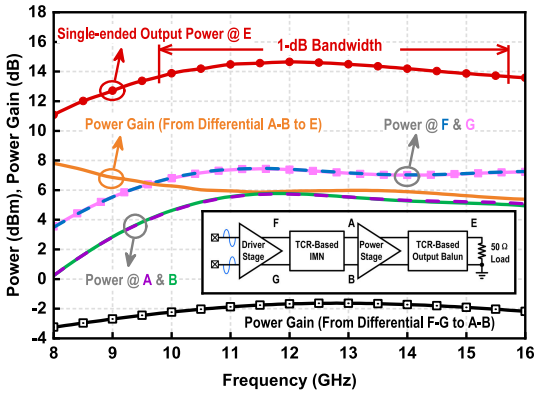


Fig. 18. Simulated power and power gain versus frequency for the cascaded two-stage PA with interstage and output matching networks [20].

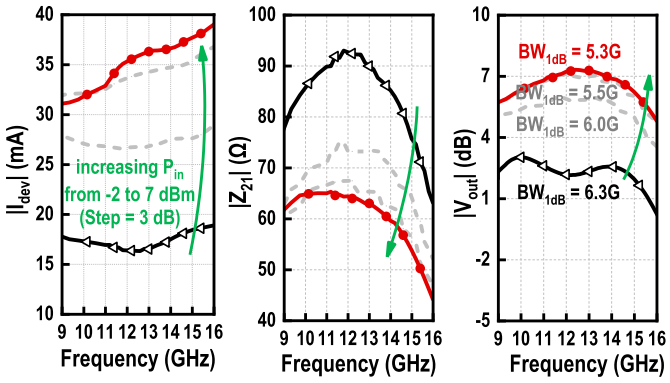


Fig. 19. Extracted $|I_{\text{dev}}|$, $|Z_{21}|$ and $|V_{\text{out}}|$ versus PA input power (P_{in}).

close to 6 GHz, thus verifying the design of the IMN. Although different from the simulation condition used in Fig. 15, there is a boosting effect on the power gain (from differential nodes A-B to E) at low frequencies due to the power response at node A/B and the compression state of the power stage. Therefore, the bandwidth of the output power at 50-Ω load is maintained and behaves the same as that shown in Fig. 15.

D. Discussions on P_{sat} Bandwidth

For broadband PA, P_{sat} bandwidth is important, especially in radar applications. Taking IMN as an example, Fig. 19 shows the simulated results as the input power level further increases. The extracted $|I_{\text{dev}}|$ shows a more significant increase at high frequencies, while $|Z_{21}|$ shows a decrease. Although the variation trends of $|I_{\text{dev}}|$ and $|Z_{21}|$ are complementary, $|V_{\text{out}}|$ bandwidth (at $P_{\text{in}} = 7$ dBm) still deteriorates compared to the synthesized results from the algorithm (at $P_{\text{in}} = -2$ dBm) because of faster decline of $|Z_{21}|$. This simulation leads to a key design process in broadband PA, which is not only to meet the requirement of center frequency, but more importantly, to design PA at the compression state. As described in the flowchart in Fig. 10, operating the active stage near $P_{1\text{dB}}$. In this way, based on the proposed algorithm, PA can obtain the required power bandwidth near $P_{1\text{dB}}$, while narrowing the gap between P_{sat} bandwidth and synthesized bandwidth.

V. Measurement Results and Discussions

Fabricated in 110-nm 1P7M CMOS process, the proposed two-stage PA occupies a core area of 0.2 mm^2. Fig. 20 shows the circuit schematic and chip micrograph, respectively. The interstage matching network and output balun of the PA are implemented based on the proposed KASM. However, the input balun is designed by using the impedance transformation [13] and conjugate matching theories, rather than the proposed KASM. The coupling coefficient k of input balun is co-designed with the parasitic resistors to achieve the required bandwidth. In PA chip measurement, the input and output RF pads are directly accessed by G-S-G probes while the DC pads are wire-bonded to a printed circuit board (PCB). Both of the driver and power stage are operated with a single 1.2-V

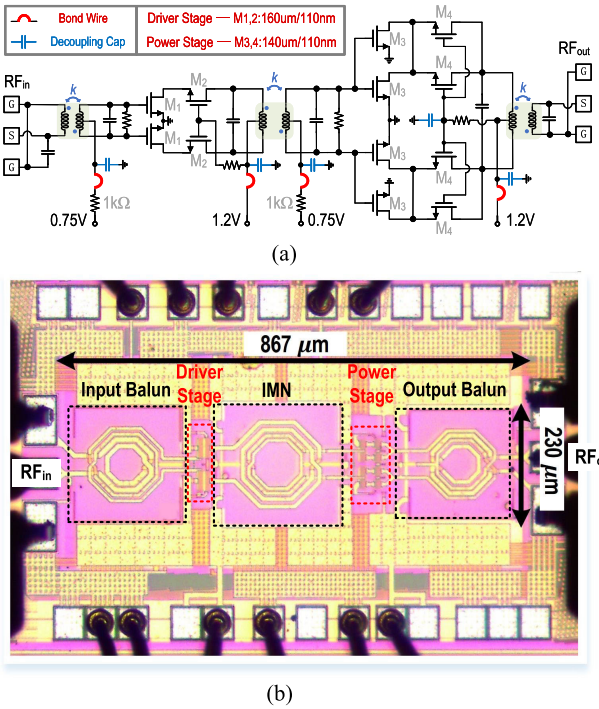


Fig. 20. (a) Schematic and (b) chip micrograph of the prototype PA [20].

supply. Each DC bias for transistor gate is to connect a 1-kΩ resistor in series on the PCB, which is used in conjunction with the carefully selected decoupling capacitors (i.e., MIM capacitors), to prevent low-frequency oscillation and ensure decent common-mode rejection ratio [27]. All DC voltages are provided by the power supply instrument of Keysight E36313A. The measured quiescent currents for the driver and power stage are 41 mA and 73 mA, respectively.

The small-signal S-parameter measurements are carried out by the vector network analyzer of Keysight N5247B. As shown in Fig. 21, the designed PA achieves a peak S_{21} of 19.5 dB with a 3-dB bandwidth ($\text{BW}_{\text{-3dB}}$) from 9.48 to 15.5 GHz (i.e., 48.2% FBW), and the input return loss is better than 10 dB in the frequency range of 10.5-16.3 GHz, showing broadband input matching. The PA is unconditionally stable during the measurement. The S_{21} is less than the simulation

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TABLE I
COMPARISON OF THE PUBLISHED BROADBAND PAS IN SILICON

Ref.	Process	BW_{-3dB} (GHz)	Supply (V)	Gain (dB)	P_{sat} (dBm)	PAE_{max} (%)	$^{a}FBW_{-3dB}$ (%)	P_{sat} FBW_{-1dB} (%)	Core Area (mm^2)	FOM_1 (%)	FOM_2 (dB·%·%/mm^2)
[28] MWTL'17	65-nm CMOS	13.5-19.0	1.2	20.7	13.9	20.0	33.8	26.4**	0.40	5.28	92
[29] TMTT'20	130-nm SiGe	22-43**	2.4	11.2	14.0	25.0	64.6**	36.1	0.14	9.03	467
[30] TMTT'21	130-nm SiGe	27.3-38.5	2.5	33.5	21.3	24.0	34.0	15.9**	0.36	3.82	121
[31] SSCL'21	28-nm CMOS	22.8-34.2	2.0	13.8	17.4	40.5	40.0	19.4	0.09	7.86	482
[18] TCAS'21	130-nm SiGe	30-41.1	3.3	25.3	22.8	27.0	30.9	20.9**	0.48	5.64	92
[9] MWTL'22	28-nm CMOS	26-39.8	N/A	19.0	17.1	25.3	42.0	38.7	0.16	9.79	488
[32] MWTL'22	180-nm CMOS	14-22	3.6	12.0	16.6	18.7	44.4	36.6**	0.56	6.84	76
[33] TCAS'23	130-nm SiGe	13-21**	4.2	21.4	30.0	25.3	47.1**	17.8	1.13	4.50	40
[34] MWTL'19	65-nm CMOS	8.0-11.5**	1.2	24.4	20.5	24.5	35.0	25.6	0.48	6.27	112
This work	110-nm CMOS	9.48-15.5	1.2	19.5	15.7	27.0	48.2	52.2	0.20	14.09	662

**Graphically estimated, $^{a}FBW_{-3dB} = BW_{-3dB}[GHz]/f_c$ [GHz], where f_c represents the center frequency of the S_{21} BW_{-3dB}
$FOM_1 = P_{sat}$ $FBW_{-1dB}[\%] \times PAE_{max}$ $FOM_2 = P_{sat}$ $FBW_{-1dB}[\%] \times FBW_{-3dB}[\%] \times$Power Gain[dB]$\times PAE_{max}$/Core Area[$mm^2$]

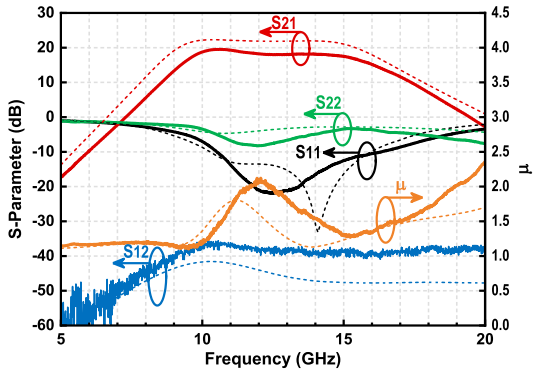


Fig. 21. Measured (solid lines) and simulated (dashed lines) results of the small-signal S-parameters.

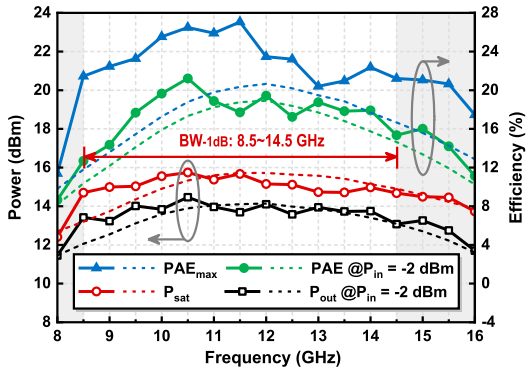


Fig. 22. Measured (solid lines) and simulated (dashed lines) results of the large-signal performance versus frequency.

results because of the imperfect grounding and the reduced quiescent currents. The latter is potentially caused by the difference between processed transistors and the simulated models.

The large-signal performance is measured using a Ceyear 4051H spectrum analyzer with excitation power provided by a Ceyear 1465H continuous wave signal generator. Fig. 22 shows P_{out} and PAE when the input power P_{in} is set to -2 dBm to obtain the desired voltage at the driver input. It can be observed that $P_{out}@P_{in} = -2$ dBm acquires a BW_{-1dB} from 8.5 to 14 GHz (i.e., 48.9% FBW) with the maximum value of 14.5 dBm, verifying the broadband design for large-signal response at the desired frequencies. The peak PAE@$P_{in} = -2$ dBm of 21.5% is achieved. Besides, the measured saturated output power (P_{sat}) versus frequency shows a 8.5-14.5-GHz BW_{-1dB} (i.e., 52.2% FBW) with a 15.7-dBm peak value at 11.5 GHz, and the measured PAE_{max} also peaks at 11.5 GHz with the value of 27%.

Fig. 23 presents the measured and simulated power and PAE performance at 12 GHz. The prototype PA achieves a 15.1-dBm P_{sat} with 23.9% PAE_{max}. The discrepancy between measured and simulation results at low-P_{in} case mainly results from the reduced currents of the active stages. As for the high-P_{in} case, the compression state of PA alleviates the impact of currents, and thereby the measured power and gain responses are substantially consistent with the simulations.

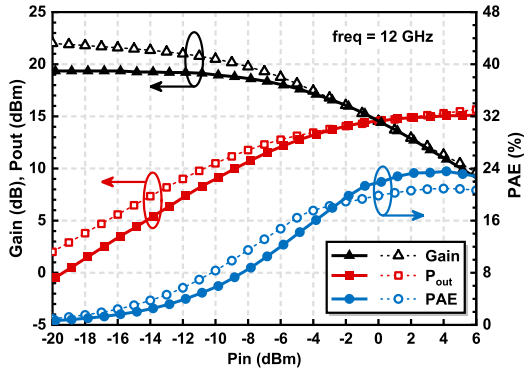


Fig. 23. Measured (solid lines) and simulated (dashed lines) results of the power and PAE performance at 12 GHz.

Table I summarizes the overall performance of the recently published broadband PAs in silicon [9], [18], [28], [29], [30], [31], [32], [33], [34]. Thanks to the proposed TCR design algorithm, the broadband PA in this work simultaneously achieves the highest FOM_1 and FOM_2, indicating the

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significant advantages in amplification bandwidth, PAE, power gain and chip cost. The notable bandwidth performance of both linear power gain and P_{sat} show the potential of the proposed algorithm for different application requirements (e.g., mixed-mode FMCW radar that necessitates broadband amplification for both short- and medium-range operations).

VI. Conclusion

This work presents a knowledge-aided automated synthesis method for broadband TCR-based PAs. The intrinsic design knowledge is derived from the detailed analysis of TCR impedances and the interaction between PA active stages and TCR networks. By combining the design knowledge with an optimization-oriented GA, the proposed method is able to efficiently synthesize the optimal TCR parameters for multi-stage PA to realize the user-defined specifications including the output power and 1-dB bandwidth. The efficient synthesis is attributed to the superiority of the GA initial population that directly generated by knowledge, which improves the algorithm convergence and reduces the iterations during GA optimization. Compared to the traditional GA, the proposed automated design method achieves about 4 times speed improvement. With the aim of 14-dBm output power and 50% 1-dB FBW, a two-stage cascode-based PA for FMCW radar application is designed with the proposed method in 110-nm CMOS process. The fabricated PA chip shows a peak P_{sat} of 15.7 dBm with 52.2% 1-dB FBW and thus validating the efficacy of the proposed synthesis method for broadband PAs.

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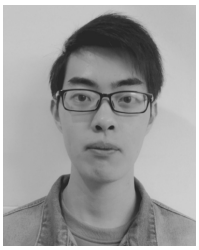
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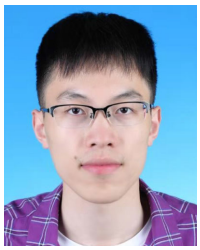
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